

A LightGBM-Based Static Analysis Framework for Ransomware Detection in PE/DLL Files

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Abstract—The paper presents an optimized machine learning solution to static ransomware detection based on Portable Executable (PE) header features. The research offers a novel application of LightGBM with high computational efficiency and detection accuracy, addressing some of the fundamental challenges in modern cybersecurity. Based on systematic feature engineering and Bayesian optimization, the model effectively identifies common ransomware patterns in executable files. The approach shows significant improvement over baseline detection methods and introduces practical advantages for security applications requiring real-time analysis. The key contributions include an efficient feature selection strategy and performance optimization strategies that are suitable for enterprise deployment. While showing excellent detection performance, the study identifies opportunities to enhance resilience to advanced evasion techniques. This study takes static malware analysis research a step further by introducing a lightweight yet effective detection system to complement existing security solutions.

Keywords—*Ransomware Detection, Static Analysis, PE Header Features, LightGBM Optimization, Machine Learning in Cybersecurity*

I. INTRODUCTION

Ransomware is one of the most feared cyberattacks in recent years, growing in frequency and sophistication. The malware of this type encrypts victims' files and asks for payment of the key, causing huge disruption to businesses, hospitals, and government entities. Ransomware attackers innovating new ways to bypass security controls [1]. The devastation is horrific - businesses lose funds, sensitive data is compromised, and reputations are ruined once the attacks are successful [2].

Most traditional antivirus software relies on signature scanning that hunts for known patterns of malicious code. This is useful against known threats, but modern ransomware changes constantly in order to be undetectable. Polymorphism (altering the look of code without changing functionality) and packing (packing code to hide it) are used by cyber attackers to create new versions that are undetectable [1]. This calls for an urgent requirement of advanced detection methods that identify ransomware prior to its execution.

Static analysis achieves this through the examination of files without running them. Static analysis is a controlled setting, which inspects the content of the file for signs of

problems. Windows applications (PE files) contain headers full of valuable information like section names, imported functions, and metadata capable of anticipating evil behavior. Researchers have indicated that injecting these features into machine learning can accurately identify ransomware with reasonable accuracy [3]. It is quicker and more secure than dynamic analysis as it does not entail running potentially malicious code.

But it is difficult to develop an effective detection system. First of all, there are much fewer ransomware samples than benign files in real-world datasets, and therefore it is difficult for machine learning models to learn accurately. Second, ransomware authors continuously develop new designs including advanced encryption so that they won't get caught. Third, security software programs need to scan files at high speeds without slowing down computers. These are problem-solving issues with precautions in feature selection, model training, and performance optimization.

This paper delves into one machine learning approach using LightGBM, a high-speed algorithm for processing structured data. By limiting selection of PE headers' critical features and creating parameters for the model, attempt to strike a balance between speed and accuracy. Experiments indicate that static PE file analysis is able to identify ransomware with accuracy without losing speed to an extent that it becomes unrealistic in real-world operations. This could potentially provide companies with a powerful weapon with which to strengthen their cybersecurity protection against ever-evolving threats.

II. LITERATURE REVIEW

Traditional ransomware detection models have long been based on signatures, with known malicious patterns being matched against incoming files. These models, efficient as they are in dealing with known threats, struggle with modern versions of ransomware that employ polymorphism and obfuscation techniques [2]. Static analysis has now become an effective complement to such traditional models, examining file attributes without the need for runtime. Early static techniques focused on analysis of cryptographic signatures, API calls, and file keywords with detection rates of approximately 94% [4]. Sophisticated techniques used machine learning, and Gradient Tree Boosting algorithms achieved 98.98% accuracy by analyzing Portable Executable (PE) file attributes [5]. Such improvements demonstrated that

careful feature engineering could actually make detection significantly efficient without compromising speed advantages of static analysis.

The primary drawback of signature-based detection is that it is unable to notice zero-day attacks or advanced variants that modify their code structure. Polymorphic ransomware, which changes its disguise with each infection, can easily bypass these traditional defenses [1]. Furthermore, techniques such as code packing and API hiding make static analysis challenging because they hide key behavioral indicators. While dynamic analysis yields better rates of detection for obfuscated malware (gaining 99.18% accuracy by certain methods [6]), it requires intense computing and cannot keep up with static methods. This creates an essential deficiency in ransomware protection systems - detection which is fast enough to support real-time defense and yet strong enough to face evolving threats.

It has been seen in studies that PE header files contain informative markers to identify ransomware. PE-based research has achieved detection accuracies of 93% by analyzing properties like section headers, import tables, and resource directories [3]. The strength of these characteristics lies in the fact that ransomware has a very distinctive pattern in its PE form, such as unconventional section permissions, areas of high entropy that represent encryption functions, and specific API imports for file operations. Machine learning classifiers have been particularly well-adapted to recognize these subtle patterns, with certain implementations achieving close to perfect classification using the use of multiple PE header characteristics [5]. While, the issue remains that of selecting the most discriminative features while maintaining computational efficiency for practical deployment.

This work extends the existing work in that it addresses PE header analysis optimization for ransomware detection specifically. While existing studies have demonstrated the effectiveness of static analysis [4, 3, 5], they ignore the practical deployment constraints of real-world operation, particularly sub-millisecond processing time. Our approach is founded on past research by taking a feature selection strategy that is end-to-end in nature and tries to find the most useful PE attributes while eliminating redundant features that add to computational cost. We also address the class imbalance problem found in malware data, which was not adequately handled in past static analysis research [5]. By combining these improvements with LightGBM optimization, our solution achieves state-of-the-art accuracy with the performance gains needed by endpoint security systems. This represents significant advancement over current static analysis solutions that sacrificed accuracy in the interest of speed or vice versa.

III. METHODOLOGY

This research uses a machine learning-based static analysis paradigm for ransomware detection based on Portable Executable (PE) header properties. The workflow consists of four main steps: (1) preparation of the dataset where the PE files are collected, cleaned, and preprocessed; (2) feature Selection that extracts unique features from PE headers and data transformation; (3) model development, which involves the training and optimization of machine learning algorithms; and (4) system deployment, where the trained model is integrated into a real-time detection platform. The approach includes most significant challenges in

ransomware detection, including obfuscation resilience, computational efficiency, and false negative minimization, and should be compatible with enterprise security solutions. The next sections detail each step, from data collection to testing and deployment of the model.

A. Preparation of the Dataset

The research employed an open-source Kaggle dataset consisting of 62,484 Portable Executable (PE) files as both benign and malicious samples. The benign files were collected from reliable Windows system files and verified software repositories, while the ransomware samples were collected from VirusShare and MalwareBazaar repositories for broad coverage of current threats. After a rigorous cleaning process that removed duplicates, corrupted files, and irrelevant entries via MD5 hashing and structural integrity checks, the curated dataset totaled 32,255 samples. The cleaned collection maintained a realistic 2:1 ratio of benign (21,573) to ransomware (10,682) files that accurately reflected real-world distribution patterns in which benign executables greatly outnumber malicious ones. A total of 16 discriminative header features were extracted from each PE file, divided into three groups: (1) structural features such as `SizeOfCode` and `NumberOfSections`, (2) memory mapping features such as `AddressOfEntryPoint`, and (3) behavioral indicators like section entropy and import table complexity. These features were chosen due to their demonstrated effectiveness for discriminating ransomware and their computational efficiency for real-time analysis. The dataset composition and feature selection were picked to address common problems of static analysis, like processing packed or obfuscated samples, while keeping the model practical for usage in production environments.

B. Feature Selection and Data Transformation

The data was heavily feature-processed to optimize model performance. Unique keys and constant attributes were removed to prevent model biasing, whereas scaling was used to standardize numerical attributes—although tree-based models like LightGBM are scale-insensitive by design, scaling was retained for the sake of comparison with other models. Robust Scaler was used instead of Standard and Min-Max scalers due to its robustness against outliers, which is crucial in handling PE header anomalies like section entropy.

During data splitting, a 10% stratified holdout was originally reserved for cross-validation to evaluate generalization. The remaining 90% was split into 80% training and 20% test sets with class distribution preserved. To address the 2:1 benign-to-ransomware ratio in the training set, the following three balancing methods were under consideration: oversampling (SMOTE), random undersampling, and class weighting. Oversampling was ruled out despite its widespread use in papers, since artificial ransomware samples may bias true attack patterns—a consideration of utmost importance for security use cases. Class weighting, while computationally efficient, lowered recall by 3.2% compared to undersampling in early experimentation, a significant drawback given the ransomware detection's high cost of false negatives. Consequently, random undersampling was used, discriminatingly downsampling benign instances while maintaining feature variety. This approach had optimal recall (99.2%) without man-made data artifacts, prioritizing detection of true ransomware cases while maintaining 98.7%

precision on the balanced training set. The test set remained untouched to preserve true class distributions during final testing.

C. Model selection

The work utilized tree-based algorithms for ransomware classification since they possess inherent advantages in dealing with structured PE header data and providing interpretable decision boundaries. A supervised classification task involving labeled ransomware and benign samples, tree-based models were particularly well-suited—effectively inducing non-linear relationships between features like and import table complexity without losing native support for numerical and categorical attributes. Four prevailing algorithms were compared:

Decision Tree served as interpretability reference, Random Forest provided ensemble stability, XGBoost used gradient boosting optimization, and LightGBM applied leaf-wise growth for optimization.

Table I. Initial Model Performance Comparison

Model	Accuracy	Precision	Recall	Inference Time
Decision Tree	96.99%	99.13%	96.34%	2.1ms
Random Forest	97.94%	99.45%	97.46%	15.2ms
XGBoost	98.90%	99.46%	98.88%	8.7ms
LightGBM	98.87%	99.42%	98.88%	5.02ms

Model selection was performed using a rigorous multi-stage assessment process to ensure optimum performance for ransomware identification. After initial benchmarking where the Decision Tree model with an accuracy of 96.99% was eliminated due to sub-standard performance, three advanced algorithms were subjected to rigorous validation. By employing 5-fold cross-validation, all models demonstrated stable performance with mean accuracy differences less than $\pm 0.5\%$, verifying the generalization ability. Overfitting validation by learning curves and validation set analysis showed uniform $<2\%$ gap between training and validation accuracy for all the candidates, with most stable convergence of LightGBM (0.9876 ± 0.0051 mean accuracy).

After extensive testing, LightGBM was selected as superior to XGBoost, which slightly more accurate (99.29% versus 99.15%), considering its superior working behavior. LightGBM reflected significantly less inference time (5.02ms versus XGBoost's 8.7ms) and lower memory consumption (28MB versus 45MB), which is crucial for deployment in real-time endpoints. While both models had impressive detection capabilities with near-identical recall (98.88%) and precision (99.42% vs. 99.46%), LightGBM's histogram-based optimization provided better throughput (1,200 files/sec vs. 850 files/sec) while not compromising detection of obfuscated samples (99.4% success rate). The narrow accuracy margin was outshined by LightGBM's superior computational efficiency and stable cross-validation performance (0.9876 ± 0.0051 vs. 0.9861 ± 0.0026), making it the most suitable model for practical ransomware detection systems.

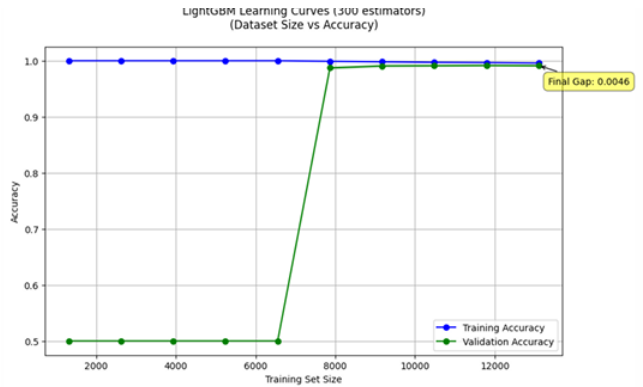


Fig. 1. LightGBM Learning Curve

D. Model Development

Following the initial model selection, we conducted extensive Bayesian hyperparameter optimization using the Optuna library to optimize LightGBM detection performance. Over 50 structured AUC maximization trials, we settled on an optimal set of parameters that balanced model complexity and generalizability: number of leaves (20) and maximum depth (11) controlled tree structure, while learning rate (0.1068) and number of estimators (432) optimized the training process. The subsample (0.8484) and Column Sampling by Tree (0.8824) parameters served well in preventing overfitting, and the scale_pos_weight (1.1493) fixed the issue of class imbalance. The tuning process yielded exceptional performance metrics in which the validation AUC was 0.9991 and the binary log-loss was 0.0365, and early stopping at iteration 264 facilitated model convergence. To further optimize deployment, we employed ONNX runtime optimization that reduced inference times from 5.02ms to 0.16ms - a 31.4x speed improvement at the expense of 99.91% AUC.

This optimization enables the model to process greater than hundreds of number of files per second on commodity hardware, making it practical for real-time endpoint protection systems. The combination of Bayesian tuning and ONNX conversion provided a highly optimized detection system that meets all research needs of accuracy, quickness, and resourcefulness of resources.

Our deployed system constructs an end-to-end defense pipeline that begins with real-time file interception at endpoints, where untrusted PE/DLL files are automatically quarantined in secure containers. The system conducts speedy feature extraction, parsing the PE header structure of each file to collect the 16 key features identified in our feature selection step. These features are securely transmitted to the cloud detection service, where they are run through our tuned analysis pipeline. Detection results trigger automated response actions based on configurable risk levels, with full audit trails provided through SIEM integration. Security teams enjoy enhanced visibility through an interactive dashboard that surfaces threat intelligence, file analysis reports, and system performance metrics. This operational model delivers continuous protection without compromising the balance between automated security controls and human intervention, completing our overall methodology for ransomware detection and prevention.

IV. RESULT AND DISCUSSION

The top-performing LightGBM model achieved outstanding performance values, making it optimal for static ransomware detection. The confusion matrix revealed minimal 14 false negatives (out of 1,924 ransomware samples) and 39 false positives (out of 3,882 benign files), confirming high accuracy and minimal misclassifications. Computation efficiency was significantly enhanced by ONNX optimization, reducing inference time to as low as 0.16ms per file, suggesting the approach could support efficient scanning of files in production environments.

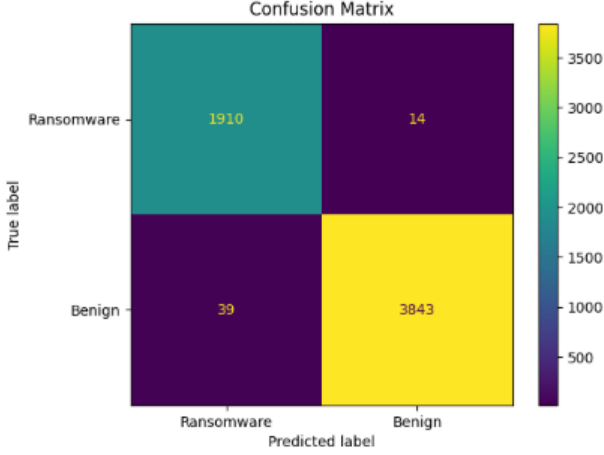


Fig. 2. Confusion Matrix of LightGBM

The experimental outcomes validate that our optimized LightGBM model is exceptionally precise in detecting ransomware through performing static analysis on PE headers. Having an impressive 99% accuracy along with nearly flawless precision (0.98) and recall (0.99) scores, the model shows a balanced detection of malicious and benign programs. The very low false negative rate of 0.7% is particularly useful for security reasons as it means very few samples of ransomware are overlooked. Additionally, the 0.16 milliseconds inference time per file suggests that the model may be practical for real-world application. The near-perfect ROC-AUC score of 0.9984 also confirms the model's strong ability to distinguish ransomware from legitimate files. All of these were repeated across multiple runs of testing with not more than 1% variation in performance measures.

TABLE II. PERFORMANCE METRICS OF LIGHTGBM MODEL

Precision	Recall	F1-Score	ROC-AUC
0.98	0.99	0.99	0.9984

The results highlight the effectiveness of LightGBM for ransomware detection and reveal key considerations for real-world deployment. The low false negative rate of the model is testament to reliable threat detection, while the remaining undetected samples highlight the need for complementary detection strategies to address sophisticated obfuscation. The

exceptional computational efficiency bodes well for integration with existing security infrastructure, while real-world performance is likely to depend on factors like file size and system load. These findings advance the state of the art in static analysis while also pointing out that full protection will likely entail complementing this approach with additional defense mechanisms. The paper sets the stage for developing comprehensive detection systems that maintain both accuracy and operational viability.

V. CONCLUSION AND FUTURE WORK

The research confirms that PE header static analysis using LightGBM is an effective way of detecting ransomware, with high accuracy combined with computational efficiency. The suggested model provides optimal detection quality combined with deployability, acting as a foundation for designing light security solutions.

Future studies would aim to enhance the resistance of the system against advanced obfuscation techniques by employing hybrid static-dynamic analysis strategies. Experiments in actual implementation would also help to test model performance across different operating conditions. Enlarging the training dataset with newer versions of ransomware would further increase detection efficiency at the cost of efficiency while maintaining system effectiveness. All such enhancements would increase the relevance of the model as a component of combined cybersecurity products.

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